

Lagzouli Riyad

Data Science Engineering Student — INSEA, Rabat

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riyad-77.github.io/riyad-portfolio

PROFILE

Third-year Data Science engineering student at INSEA. Expertise in statistical modeling (ML/Deep Learning), deterministic expert systems, and modern data architectures (ETL/Lakehouse). Hands-on proficiency in Python, SQL, Airflow, dbt, and Docker, with proven experience building real-time microservices. Seeking a final-year internship (PFE) starting **February 1, 2027** to design, optimize, and deploy high-impact data-driven solutions.

TECHNICAL SKILLS

Machine Learning & Modeling: Random Forest, XGBoost, Regression, Classification, Clustering, SMOTE

Deep Learning & NLP: Transformers (BERT), Word Embeddings, Neural Networks, TF-IDF

Time Series & Statistics: Box-Jenkins Methodology, Stationarity Tests (ADF, KPSS), Statistical Inference

Decision Support & Expert Systems: Deterministic expert systems, rule-based inference engines, formal logic, model explainability

Data Engineering & Architecture: Apache Airflow, dbt, DuckDB, PostgreSQL, PostGIS, ETL/ELT Pipelines, Medallion Architecture

Databases & Query Optimization: Advanced SQL (Relational Modeling, Analytical Queries, Indexing), RDBMS

Development, Backend & DevOps: FastAPI (Latency $<$ 50ms), Docker, Git/GitHub, Python, R

Business Intelligence & Reporting: Power BI, DAX Modeling, Interactive Executive Dashboard Design

PROFESSIONAL EXPERIENCE

Data Science & Engineering Intern June 15 – September 11, 2026

Bifolia

- Architected a deterministic expert system for mission-critical decision support, guaranteeing 100% explainability and zero hallucination
- Orchestrated automated Airflow ETL pipelines, satellite data ingestion, and spatial modeling (PostgreSQL/PostGIS) under Docker
- Built a high-performance FastAPI backend for real-time inference of predictive models with sub-50ms latency
- GitHub: github.com/riyad-77/SmartFellah

Statistical Analysis Discovery Internship July 2025

High Commission for Planning (HCP) — Regional Directorate of Fez-Meknes

- Analyzed the National Survey on the Informal Sector to identify structural macroeconomic patterns and production unit behaviors
- Explored raw micro-datasets to ensure data reliability and consistency

Appointment Scheduling Agent January – August 2023

NL CALL

- Managed outbound call campaigns, client qualification, and operational appointment booking

PROJECTS

Modern Data Lakehouse — Bank Reviews Analysis 2025 – 2026

Python (Selenium) · DuckDB · dbt · Airflow · Power BI

- End-to-end Medallion pipeline: web scraping banking reviews, SQL transformations, automated data quality tests, and dashboards
- GitHub: github.com/riyad-77/Bank-Reviews-Lakehouse

Credit Card Fraud Detection 2025 – 2026

Python · Scikit-learn · Random Forest · XGBoost · SMOTE

- Machine learning pipeline comparing ensemble models (Random Forest, XGBoost) and handling extreme class imbalance with SMOTE
- GitHub: github.com/riyad-77/Credit-card-fraud-detection

Time Series Analysis — Business Bankruptcies 2024 – 2025

R · Box-Jenkins (ARIMA/SARIMA) · Statistics

- Modeling quarterly business bankruptcies in Paris, stationarity testing (ADF, KPSS), and out-of-sample risk forecasting
- GitHub: github.com/riyad-77/Series-chronologiques

Stock Market Risk Analysis — Data Viz 2025 – 2026

Power BI · Yahoo Finance · DAX

- Risk monitoring and analysis dashboard tracking market volatility and financial indicators
- GitHub: github.com/riyad-77/Power-BI

NLP Spam Detection 2025 – 2026

Python · TF-IDF · BERT (Transformers) · scikit-learn

- Text classification comparing TF-IDF vs BERT embeddings for textual anomaly and fraud detection
- GitHub: github.com/riyad-77/NLP-Spam-Detection

EDUCATION

Engineering Degree in Data Science 2024 – Present

National Institute of Statistics and Applied Economics (INSEA), Rabat

University Diploma (DEUST) — Mathematics, CS, Physics 2022 – 2024

Faculty of Science and Technology, Fez (High School Diploma in Physical Sciences in 2022)

CERTIFICATIONS

IBM: Python for Data Science
NVIDIA: Fundamentals of Deep Learning
DataCamp: Supervised Learning (scikit-learn)
DataCamp: Unsupervised Learning in Python
DataCamp: Advanced Git

LEADERSHIP & ACTIVITIES

President of INSEA IT Club: LLM/Web workshops, negotiated 100 DataCamp licenses, NVIDIA workshop.

LANGUAGES

Arabic: Native
French: Fluent / Bilingual
English: Intermediate

INTERESTS

Travel · Sports · Music